Limit: Formula Sheet

Existence

LHL: $\lim_{h\to 0} f(a-h)$

RHL: $\lim_{h\to 0} f(a+h)$

Exists: LHL = RHL = finite

Indeterminate:

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty-\infty, 0\cdot\infty, 1^\infty, 0^0, \infty^0$$

Algebra and rationalisation

Sum/product: $\lim(f\pm g) = l\pm m$, $\lim fg = lm$

Quotient: $\lim \frac{f}{g} = \frac{l}{m}$ $(m\neq 0)$

Square:

$$a-b = \frac{a^2-b^2}{a+b}$$

Cube:

$$a-b = \frac{a^3-b^3}{a^2+ab+b^2}$$

Substitutions

$x\to a$: $x = a+h$

$x\to\infty$: $x = \frac{1}{t}$, $t\to 0^+$

$x\to -\infty$: $x = -t$, $\sqrt{x^2} = -x$

Infinity/infinity

Degrees: $\frac{a_0}{b_0}$ $(m=n)$, 0 $(m<n)$, $\pm\infty$ $(m>n)$

Growth: $\ln x \ll x^n \ll e^x$

Near 0^+: $x\ln x\to 0$

Standard limits

Power:

$$\lim_{x\to a}\frac{x^n-a^n}{x-a} = na^{n-1}$$

Binomial:

$$\lim_{x\to 0}\frac{(1+x)^n-1}{x} = n$$

Trig:

$$\frac{\sin x}{x}, \frac{\tan x}{x}, \frac{\sin^{-1}x}{x}, \frac{\tan^{-1}x}{x}\to 1$$

Cosine:

$$\frac{1-\cos x}{x^2}\to\frac{1}{2}$$

Exponential: $\frac{e^x-1}{x}\to 1$, $\frac{a^x-1}{x}\to\ln a$

Logarithm:

$$\frac{\ln(1+x)}{x}\to 1$$

Oscillating: $x^p\sin\frac{1}{x}\to 0$ $(p>0)$

Replacements ($s\to 0$)

Trig:

$$\sin s, \tan s, \sin^{-1}s, \tan^{-1}s \to s$$

Cosine: $1-\cos s\to\frac{s^2}{2}$

Exp/log: $e^s-1\to s$, $a^s-1\to s\ln a$, $\ln(1+s)\to s$

Power: $(1+s)^n-1\to ns$; $S\to 1$: $S^n-1\to n(S-1)$, $\ln S\to S-1$

Exponential forms

1^∞: $\lim f^g = e^{\lim g(f-1)}$

General: $\lim f^g = e^{\lim g\ln f}$

Built-in:

$$\lim_{n\to\infty} x^n = 0\ (|x|<1),\ 1\ (x=1),\ \infty\ (x>1)$$

Expansions

$$\sin x = x-\frac{x^3}{6}+\frac{x^5}{120}$$

$$\cos x = 1-\frac{x^2}{2}+\frac{x^4}{24}$$

$$\tan x = x+\frac{x^3}{3}+\frac{2x^5}{15}$$

$$e^x = 1+x+\frac{x^2}{2}+\frac{x^3}{6}$$

$$\ln(1+x) = x-\frac{x^2}{2}+\frac{x^3}{3}$$

$$\tan^{-1}x = x-\frac{x^3}{3}+\frac{x^5}{5}$$

$$\sin^{-1}x = x+\frac{x^3}{6}+\frac{3x^5}{40}$$

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2}x^2$$

Four standard: $\frac{x-\sin x}{x^3}\to\frac{1}{6}$, $\frac{\tan x-x}{x^3}\to\frac{1}{3}$

$\frac{e^x-1-x}{x^2}\to\frac{1}{2}$, $\frac{\ln(1+x)-x}{x^2}\to -\frac{1}{2}$

Special signs, sandwich, derivative

Near integer n: $[x] = n-1$ / n, $\{x\}\to 1$ / 0

Side: $\frac{\sin x}{x}\to 1^-$, $\frac{\tan x}{x}\to 1^+$, $\frac{\sin^{-1}x}{x}\to 1^+$, $\frac{\tan^{-1}x}{x}\to 1^-$

Sandwich: $g\le f\le h$, $\lim g = \lim h = l \Rightarrow \lim f = l$

GIF bound: $x-1<[x]\le x$

Derivative:

$$f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

Second:

$$f''(x) = \lim_{h\to 0}\frac{f(x+2h)-2f(x+h)+f(x)}{h^2}$$